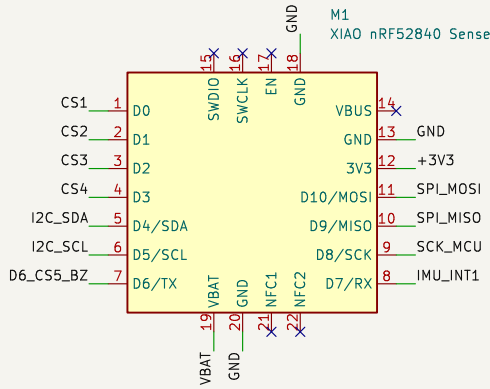
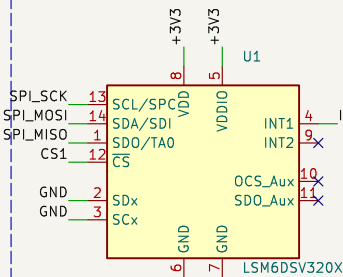


MCU + radio · XIAO nRF52840 Sense

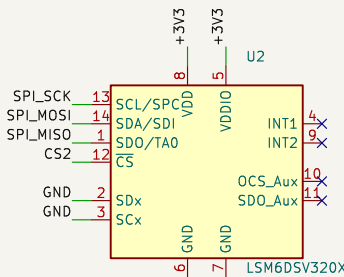


D0–D10 all used. D6 goes to JP1 (U5 chip-select or buzzer).
VBUS, SWD, EN and NFC pads left unconnected.
Firmware: P1.11 (D6) high drive for the buzzer;
charge current 100 mA (P0.13 low).

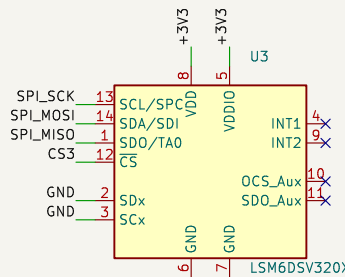
Motion sensors · 5x LSM6DSV320X on shared SPI (sensor face, all same orientation)



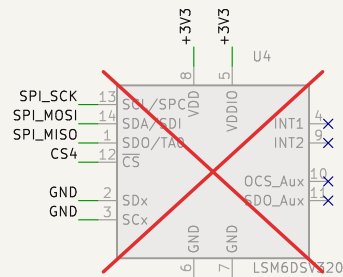
Center (0, 0)



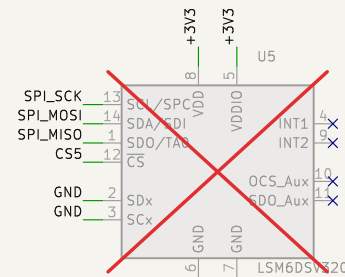
Pair A +X (+20.00, 0)



Pair A -X (-20.00, 0)



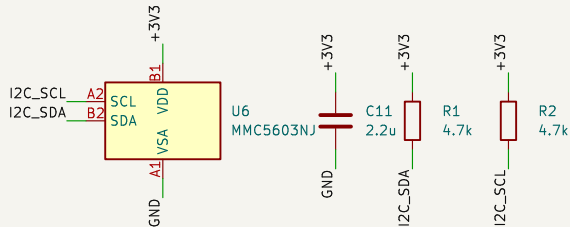
Pair B +Y (0, +20.00) – not fitted



Pair B -Y (0, -20.00) – not fitted

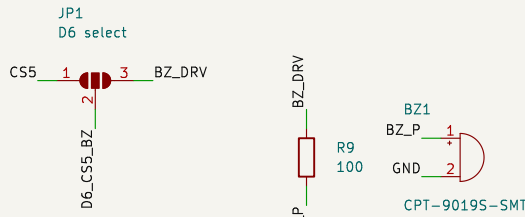
Pins: 1 SDO -> MISO · 2 SDx, 3 SCx -> GND · 4 INT1 -> D7 (U1 only) · 5 VDDIO, 8 VDD -> 3V3 + 100 nF each · 6, 7 GND
9 INT2, 10 OCS_Aux, 11 SDO_Aux: soldered, not connected (datasheet DS14623 Table 2) · 12 CS · 13 SCL/SPC -> SCK · 14 SDA/SDI -> MOSI
Firmware: first SPI write to every chip sets I2C_I3C_disable (IF_CFG 03h). High-g +/-256 g at 1.92 kHz; gyro +/-4000 dps.
U4, U5, C7–C10, R7, R8 are DNP (pair B, fitted later).

Magnetometer · MMC5603NJ (sensor face, (0, -24) mm)



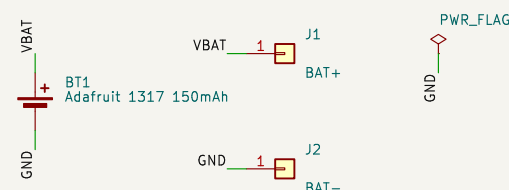
I2C address 0x30. Keep >= 10 mm from XIAO, battery and any current-carrying trace on either face. Footprint: verify ball pitch vs MEMSIC drawing.

Finder buzzer · CUI CPT-9019S on D6 via JP1



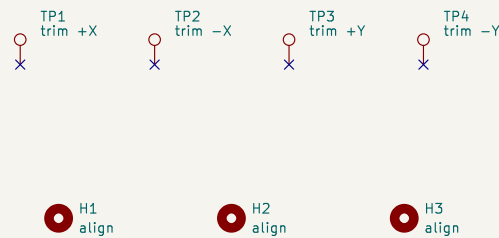
Ship JP1 bridged 2–3 (buzzer) while U5 is not fitted.
Drive D6 with a 4 kHz PWM square wave.

Power · Adafruit 1317 LiPo, 150 mAh



J1/J2 at (-18.4, +/-3) mm on the XIAO face, routed to XIAO VBAT (pad 19) / GND (pad 20). Battery bonded, centered at (-6.54, 0) mm to balance the XIAO.

Mechanical · trim pads and alignment holes



Trim pads: -Y pad sits opposite the buzzer; others for fine balance.
Holes locate the board on the pod's pins.

Board notes

56 mm round, 4 layers, 0.8 mm (0.6 mm or cutouts to save weight). Solid GND plane under the sensors.
Sensor face (machine assembled): U1–U6, C1–C11, R1–R9, JP1 · XIAO face (hand-soldered): M1, BZ1, C12, J1/J2, TP1–TP4, battery.
U1 exactly at board center; U2/U3 centers at (+/-20.00, 0); U4/U5 at (0, +/-20.00). Measure to package center.
Every off-center mass mirrored or trimmed. Magnetometer clear of XIAO, battery, buzzer and current traces.
Source of truth for positions: hardware/placement.csv · pins: hardware/pinmap.csv · overview: hardware/overview.html

Generated from hardware/kicad/gen – edit freely in KiCad 9
56 mm round · spin pair r = 20.00 mm · build: U1 + U2 + U3
Step 1: sensor board with XIAO nRF52840 Sense soldered on

Messiah University · David Whitney

Sheet: /
File: disctracker_sensor.kicad_sch

Title: DiscTracker sensor board

Size: A3 Date: 2026-10-08
KiCad E.D.A. 9.0.9

Rev: 0.1
Id: 1/1